
Effluent Discharge Compliance Standard — Bioethanol Plant (excerpt)

Supplied grounding document for capstone groups (e.g. an effluent-compliance agent) that don't have their own. Covers the site's treated water discharge consent limits.

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1. Purpose & scope

Defines the site's discharge consent limits for treated effluent leaving water treatment plant W-301, and the monitoring and reporting steps required to stay in compliance. Applies to utilities operators, shift engineers, and environmental compliance staff.

2. Discharge consent limits (final effluent, before release)

Parameter	Consent limit	Alarm (internal, tighter than consent)	Notes
pH	6.0–9.0	6.5–8.5	Continuous monitor at discharge point
BOD ₅	≤ 30 mg/L	> 20 mg/L	Weekly composite sample
COD	≤ 125 mg/L	> 100 mg/L	Weekly composite sample
Suspended solids	≤ 35 mg/L	> 25 mg/L	Daily grab sample
Temperature	≤ 30 °C	> 28 °C	Continuous monitor
Ethanol (residual)	≤ 5 mg/L	> 3 mg/L	Relevant given stillage/slops routing (see Waste Handling SOP)

The internal alarm bands are intentionally tighter than the legal consent limits, so that a deviation can be caught and corrected before the site is at risk of a consent breach.

3. Monitoring & recording

- Continuous parameters (pH, temperature) are logged automatically; a sustained excursion beyond the internal alarm band for more than 30 minutes must be logged as an event.
- Composite and grab samples are taken per the schedule in §2 and results entered in the environmental log within the same shift.
- Any known process upset that could affect effluent quality (e.g. a large off-spec stillage or slops routing event — see Waste & By-product Handling SOP) should be cross-referenced against the next effluent sample.

4. What counts as a reportable event

- Any single sample or continuous reading **exceeding the consent limit** (not just the internal alarm) is a potential reportable event to the regulator.
- Two or more consecutive alarm-band excursions on the same parameter within a week should be escalated even if the consent limit itself was not breached.

5. Escalation & human sign-off

- An assistant may flag a reading against the tables in §2, explain the likely process link (citing this standard and related SOPs), and draft a compliance log entry — but it must **not** decide whether an event is reportable to the regulator, and must **not** submit any external report.
- Any suspected consent breach must be escalated immediately to the environmental/compliance lead; regulatory notification timelines are time-critical and are the compliance lead's responsibility, not an automated system's.